

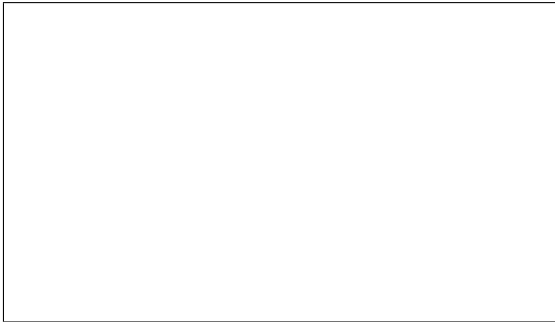
COMP5214 Assignment 3: Generative Modeling

1 Generative Adversarial Networks (60 points)

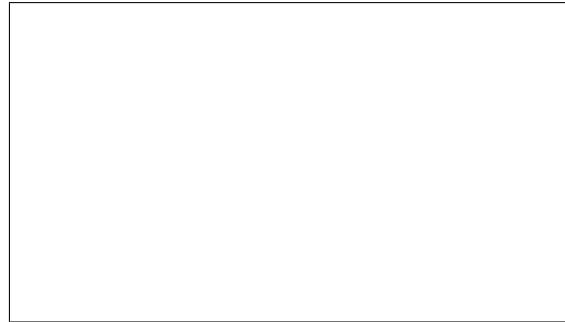
Question 1: Vanilla GAN

Solution:

Vanilla GAN FID: **[FILL IN AFTER TRAINING]**



(a) Vanilla GAN Samples



(b) Vanilla GAN Latent Space Interpolations

Figure 1: Vanilla GAN results on CUB-200-2011 (32×32).

Question 2: LS-GAN

Solution:

LS-GAN FID: **[FILL IN AFTER TRAINING]**



(a) LS-GAN Samples



(b) LS-GAN Latent Space Interpolations

Figure 2: LS-GAN results on CUB-200-2011 (32×32).

Question 3: WGAN-GP

Solution:

WGAN-GP FID: [FILL IN AFTER TRAINING]



(a) WGAN-GP Samples



(b) WGAN-GP Latent Space Interpolations

Figure 3: WGAN-GP results on CUB-200-2011 (32×32).

2 Variational Autoencoders (40 points)

Question 1: Autoencoder

Solution:



(a) Loss (size 16)



(b) Loss (size 128)



(c) Loss (size 1024)



(d) Reconstructions (size 16)



(e) Reconstructions (size 128)



(f) Reconstructions (size 1024)

Figure 4: Autoencoder reconstruction loss curves and reconstructions for different latent sizes on CIFAR-10.

Question 2: VAE with β tuning

Solution:

β^* value: [FILL IN BEST β]

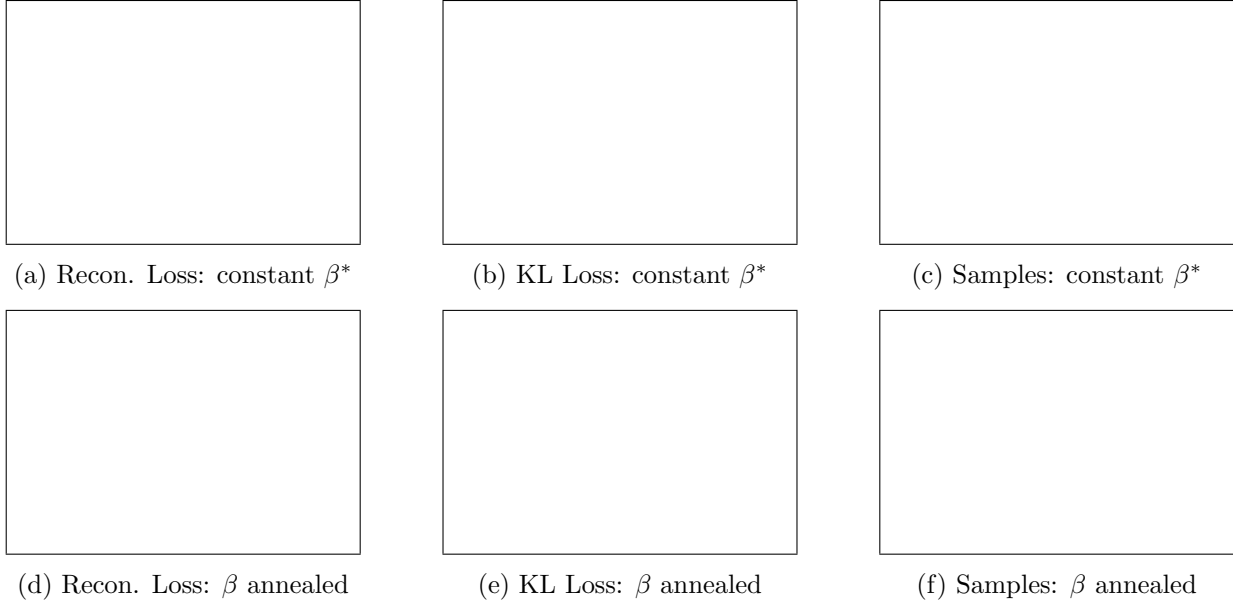


Figure 5: VAE results with constant β^* (top row) and linearly annealed β (bottom row).

3 Diffusion Models (40 points)

Question 1: DDPM and DDIM Samples

Solution:

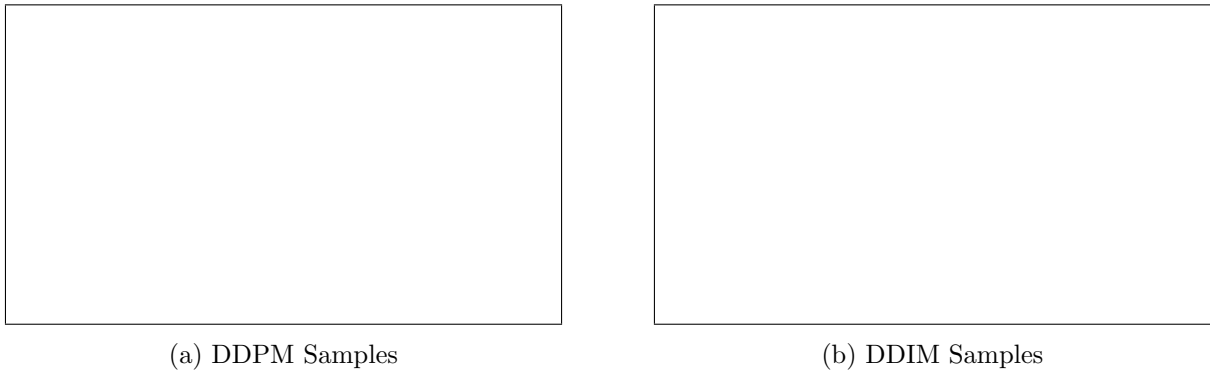


Figure 6: Generated samples from DDPM and DDIM on CIFAR-10.

Question 2: FID Scores

Solution:

- DDPM FID: **[FILL IN]**
- DDIM FID: **[FILL IN]**